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Fault Diagnosis of Brushless DC Motors Using Built-in Hall Sensors

Omid Zandi, Javad Poshtan

***Abstract*- In this paper a method for fault detection of Brushless Direct Current (BLDC) motors is proposed simply based on hall sensors that exist in most BLDC motors. Two important faults in these motors are bearing faults and demagnetization fault. In the first step the effects of bearing faults on the BLDC motor speed is studied. Then, using the Discrete Wavelet Transform (DWT) and the Continuous Wavelet Transform (CWT), a residual signal is extracted from the speed signal and analyzed in the time-frequency domain. It will be shown that the residual signal contains the main features of bearing faults such as bearing fault frequencies. In the second step the effects of the demagnetization fault on the hall sensors outputs are studied. It is shown that the demagnetization fault leads to impulsive oscillations in the duty cycle signal of the hall sensors outputs. Using the kurtosis index of the duty cycle signal, the rotor condition is classified into either a healthy condition or a demagnetized condition. In the third step, using an experimental setup, the efficiency of the proposed methods is demonstrated. The proposed algorithms are simple and only use hall sensors for fault detection, and hence lead to a simple and cheaper fault diagnosis system.**

***Index Terms*—BLDC motor, Bearing fault detection, Residual signal, Demagnetization fault detection, Discrete wavelet transform, Continuous wavelet transform**

I. Introduction

In recent years, BLDC motors have gained significant attention due to their great advantages in industry, and hence it is predicted that DC motors and induction motors are replaced with this type of motors [1]. Some of the BLDC motor advantages include higher efficiency, higher power factor, better speed-torque characteristics, higher power-to-volume ratio, and less need to maintenance [2]. Despite the common DC motors, in motors the rotor is permanent magnet, and a three-phase winding exists in the stator. Hence, there is no need for mechanical commutation using brushes, and it is replaced with electronically commutation using hall sensors. In Fig. 1 the overall view of a BLDC motor speed control system is shown.

In most BLDC motors, an internal loop exists for electronic commutation. In this loop the relative position of the rotor is determined using hall sensors. Then, based on the rotor position and according to a predetermined sequence, two windings of the stator windings are connected in series using power transistors. Consequently, the current flows through them, while the third winding remains open-circuit. Besides, hall sensors can be used for speed measurement. Therefore, there is no need for encoders or tachometers for speed measurement in these motors [3]. Two significant flaws in BLDC motors are higher cost and more complex speed control. Therefore, recently this type of motors is used in high technology

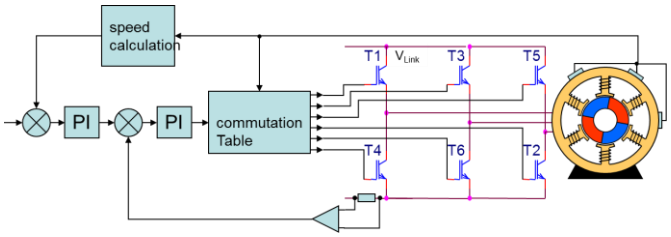


Fig. 1. BLDC motor speed control block diagram [3]

applications such as aerospace and electrical vehicles. Hence in such crucial applications fault diagnosis of BLDC motors is very crucial [4]. Also, using outer loops consisting of PI controllers, the speed and current of the motor are adjusted.

In BLDC motors, as in other motors, various faults can occur. Two important faults in these motors are bearing faults and demagnetization fault [5]. This is because bearings are used for force transmission, also the efficiency of BLDC motors severely depends on the health of the rotor magnets [2]. Therefore, this paper focuses on fault detection of these two types of BLDC motor faults.

Most existing methods for fault detection of rotating machines require special devices such as accelerometers, thermometers, acoustic sensors or current sensors [6-9]. Among these devices, accelerometers are frequently used, because each fault has a unique effect on the frequency spectrum of machine vibrations. In small motors with low level vibrations, accelerometers are not normally used due to their high costs [10]. In recent years, as a cheaper alternative, current signals are used for fault detection, and various valuable techniques based on current signals have been presented [11-13]. However, currant-based methods may not be suitable for early fault detection because only when the severity of fault is high enough, appreciable symptoms appear in the frequency spectrum of stator current signals [14].

Another important point is that, in most practical conditions, due to the time-varying operation of motors, the vibration or current signal is a non-linear or non-stationary signal. Analysis of non-linear or non-stationary signals is more complicated compared with the linear and stationary signals. For example, the well-known FFT algorithm cannot be used for analyzing these signals [15]. Hence, numerous different algorithms have been presented for fault detection of rotatory machines under non-linear or non-stationary conditions [16-19]. Some drawbacks of these algorithms include their complexity and the need for redundant (external) sensors. Furthermore, the authors found very few papers in which the proposed algorithms are not based on vibration or current signals. For example, in [20] a rotor speed-based bearing fault diagnosis algorithm has been presented, wherein by measuring motor speed using an encoder,

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the bearing defects were identified. This algorithm only detects bearing faults and needs an encoder for measuring the motor speed.

In this paper, as a cheaper and simpler alternative method with respect to vibration-based or current-based methods, fault detection of BLDC motors using hall sensors is presented. These sensors exist in most BLDC motors for electronic commutation. Therefore, since there is no need for redundant sensors such as accelerometers or current sensors, the proposed algorithms can be considered as a sensorless fault detection algorithm. It will be experimentally shown that the proposed algorithms can detect both bearing and demagnetization faults quite reliably under non-stationary operation of the BLDC motor. Moreover, they need far less computations compared with other non-stationary and non-linear fault detection strategies. The remaining part of the paper is organized as follows. Section II includes the bearing fault detection using hall sensors. Then, the demagnetization fault detection using hall sensors is presented in Section III. In Section IV, the experimental results are presented, and finally, conclusions are presented in section V.

II. BEARING FAULT DETECTION USING HALL SENSORS

A. *Effects of Bearing fault on BLDC motor speed*

The most popular types of bearings in industrial applications are ball bearings [20]. Therefore, this paper focuses on fault detection of ball bearings using hall sensors in BLDC motors. Three main important faults in these bearings include inner-race fault(IRF), outer-race fault(ORF) and ball-bearing fault(BBF). These faults have been shown in Fig. 2.

Each of these faults has a special effect on the BLDC motor torque, in that a harmonic with specific frequency will appear in the motor torque signal, as given below [21]:

$$f_c = \begin{cases} f_{out} = \dfrac{N_b}{2}\left(1 - \dfrac{d_b}{d_p}\cos(\beta)\right) f_r \\ f_{in} = \dfrac{N_b}{2}\left(1 + \dfrac{d_b}{d_p}\cos(\beta)\right) f_r \\ f_{ball} = \dfrac{d_b}{d_p}\left(1 - \left(\dfrac{d_b}{d_p}\cos(\beta)\right)^2\right) f_r \end{cases} \quad (1)$$

Where f_{out}, f_{in}, and f_{ball} are the harmonic frequencies at the ORF, IRF, and BBF faults respectively. Also d_b is the ball diameter, d_p is the pitch ball diameter, N_b is the number of balls, β is the ball contact angle, and f_r is the BLDC speed frequency. Considering only the first harmonic, the BLDC motor torque can be written as [22]:

$$T_L = T_0 + T_c \cos(\omega_c t), \qquad \omega_c = 2\pi f_c \quad (2)$$

where T_L is the load torque, T_0 is the constant load torque, T_c is the harmonic amplitude whose value depends on the severity of the fault, and ω_c is the fault frequency in radian per seconds. T_0 is actually the load torque in healthy condition, and it is supposed to be a constant torque in a constant load condition or a low frequency torque signal when the load torque varies smoothly. The mechanical equation of the BLDC motor is given by:

$$T(t) = T_m - T_L = J\frac{d\omega_r}{dt} \quad (3)$$

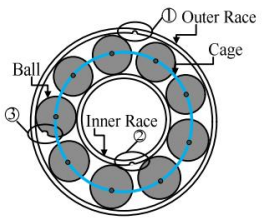


Fig. 2. Ball bearing structure and related faults [20]

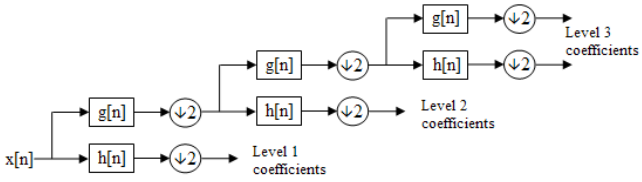


Fig. 3. Mallat algorithm for fast implementation of the DWT [25]

where T_m is the net electromechanical torque produced by the motor, J is the total inertia of the rotor and ω_r is the motor speed. Equation (3) can be rewritten as:

$$\omega_r = \frac{1}{J}\int (T_m - T_L)dt \quad (4)$$

In the steady-state operation, T_m is equal to T_0, and hence equation (4) is simplified as:

$$\omega_r = \frac{1}{J}\int (-T_c \cos(\omega_c t))dt = \omega_{r0} - \frac{T_c}{J\omega_c}\sin(\omega_c t) \quad (5)$$

Therefore, in the steady-state operation, the BLDC motor speed consists of a constant term ω_{r0} (steady-state speed) and an oscillatory term known as the residual signal. The amplitude of the residual signal depends on the severity of the bearing faults, and its frequency depends on the bearing fault type. Replacing ω_c in equation (5) from equation (1) yields:

$$\omega_r = \begin{cases} \omega_{r,out} = \omega_{r0} - \dfrac{T_c}{2\pi J f_{out}}\sin(2\pi f_{out} t) \\ \omega_{r,in} = \omega_{r0} - \dfrac{T_c}{2\pi J f_{in}}\sin(2\pi f_{in} t) \\ \omega_{r,ball} = \omega_{r0} - \dfrac{T_c}{2\pi J f_{ball}}\sin(2\pi f_{ball} t) \end{cases} \quad (6)$$

where $\omega_{r,out}$, $\omega_{r,in}$ and $\omega_{r,ball}$ are the motor speed in the outer-race, inner-race and ball-bearing fault conditions respectively.

Therefore, according to the equation (6), in case a bearing fault happens, the BLDC motor speed (at the steady-state operation) consists of a constant term and an oscillatory term whose frequency depends on the bearing fault type. Although it seems that the fault frequencies may be identified using the FFT analysis, it is important to note that the frequency-domain analysis of the speed signal is not effective because the amplitude of the residual signal (in practical applications) is very small, and the frequency peaks related to a fault are merged with the noise frequency contents, and hence, the separation between residual frequency peaks and noise frequency contents becomes an impossible task.

To solve this problem, a residual signal generator using Discrete Wavelet Transform (DWT) is proposed. The DWT algorithm can separate the low-frequency components and

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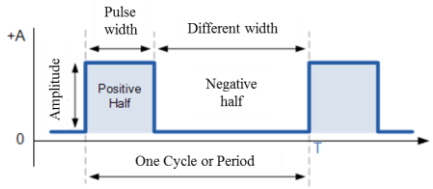


Fig. 4. BLDC motor rotor and stator structure

high-frequency components of the speed signal. Since the frequency of the residual signal compared with the frequency of the steady-state speed(ω_{r0}) is very low, the high frequency components of the DWT are related to the residual signal. Then, using Continuous Wavelet Transform (CWT), time-frequency analysis of the generated residual signal is carried out. In conditions when the motor speed in the steady-state operation is constant, Fast Fourier Transform (FFT) can also be used. In this paper however, non-stationary condition is considered wherein the motor speed reference command can be changed smoothly with respect to time.

B. Continuous Wavelet Transform

Wavelet analysis is one of the most effective approaches to analyzing non-stationary and non-linear time-domain signals. In this algorithm, a signal is studied at several diverse scales or resolutions: a large window is used to examine the low-frequency parts of the signal and a small window is used to examine the high-frequency parts of the signal. Note that studying the low-frequency and high-frequency contents of the original signal, is done simultaneously [21]. Therefore, the wavelet analysis includes a multi-resolution or multi-scale view on the signal. The wavelet analysis is established using a basis functions called wavelet (like sine and cosine functions in the FFT algorithm). Wavelet functions are designed so that they suit non-linear and non-stationary signal analysis [4]. The wavelet transform decomposes time-domain signals using time-scaled and time-shifted wavelets as in equation (7) [22].

$$Wf(t,m) = < s, \psi_{t,m} > = \int_{-\infty}^{+\infty} s(\tau) \frac{1}{\sqrt{m}} \psi^*(\frac{\tau - t}{m}) d\tau \qquad (7)$$

where t denotes time shift, m denotes time scale, and ψ is the wavelet function. The energy density of the time-domain signal, $P_W f$, is given by:

$$P_W f(t,\zeta) = |Wf(t,m)|^2 \qquad (8)$$

where $\zeta = \frac{\eta}{m}$ is the center frequency of the scaled wavelet and η is the center frequency of the original basis wavelet function. The energy density function is often plotted on a graph so-called the scalogram.

C. Discrete Wavelet Transform

Essentially, Discrete Wavelet Transform (DWT) is similar to the CWT but wavelet functions are discretized [22]. Besides, numerous algorithms, Mallat algorithm for instance, have been presented for fast calculation of the wavelet transform coefficients [23]. The Mallat algorithm utilizes the decomposition of the wavelet transform in terms of low pass (averaging) filters and high pass (differencing) filters as depicted in Fig. 3. Where $g(n)$ and $h(n)$ are the low-pass and high-pass filters which (at the first step) decompose the time-domain signal into low and high frequency contents. Low-frequency contents are known as the approximated signal, and high-frequency contents are known as the detailed signal. In the next steps, the approximated signal is again decomposed into a new detailed and approximated signal using same filters.

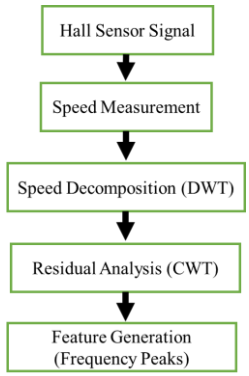


Fig. 5. BLDC motor Bearing fault detection flowchart

As mentioned earlier, the faulty speed signal at the steady-state operation includes a low-frequency signal and a residual signal with high-frequency contents. On the other hand, the detailed signal in the DWT decomposition algorithm includes the high-frequency component of the time-domain signal. Therefore, using the DWT as a signal decomposer, the residual signal is extracted from the speed signal, and in the next step the CWT is used for time-frequency analysis of the residual signal.

D. Speed Measurement Using Hall Sensors

In BLDC motors, three hall sensors are used for estimating the relative position of rotor. These sensors are placed inside the motor such that the output signals have a 120-degree electrical angle with respect to each other. When the north pole of the rotor magnets moves across the hall sensor, the output signal becomes low level, and when the south pole of the rotor magnets moves across the hall sensor, the output signal becomes high level. Therefore, the output of the hall sensors due to the continuous rotation of the rotor, forms a squared wave as in Fig. 4.

The motor speed is proportional to the frequency of the squared wave signal. Therefore, by measuring the hall sensor frequencies, the motor speed can be obtained as in equation (9):

$$\omega_r = 2\pi \times \frac{f_{hall}}{2p} \qquad (9)$$

Where p is the number of rotor poles, f_{hall} is the hall sensor output frequency, and ω_r is the motor speed at radian per seconds. Also the hall signal frequency can be measured by calculating the elapsed time between two successive rising edge or falling edge in Fig. 4. The inverse of the elapsed time is equal to the hall signal frequency. Finally, the flowchart of the proposed bearing fault detection is presented in Fig. 5.

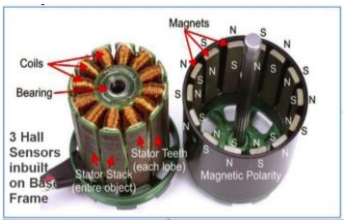


Fig. 6. BLDC motor Bearing fault detection flowchart

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III. DEMAGNETIZATION FAULT DETECTION USING HALL SENSORS

In case of a demagnetization fault, one of the magnet bars in the rotor is completely or partially demagnetized. Initially, a locally demagnetization occurs, and then, due to mechanical stresses, the magnet bar is crashed and finally a complete demagnetization occurs. In this paper, it is assumed that a complete demagnetization has occurred. Fig. 6 shows the magnetic bars in a BLDC rotor. The north and south poles of the magnetic bars in the rotor have been successively placed toward the inside of the rotor.

In a full revolution of a 12-pole rotor, there are 12 changes in the hall sensor output at healthy condition. Fig. 7 shows phase-a hall sensor output in a full revolution of the rotor. mechanical angle of 150 to 180 degrees, is completely demagnetized. In this case, when this magnet bar moves across a hall sensor, the hall sensor output remains unchanged in the previous situation, as shown in Fig. 8.

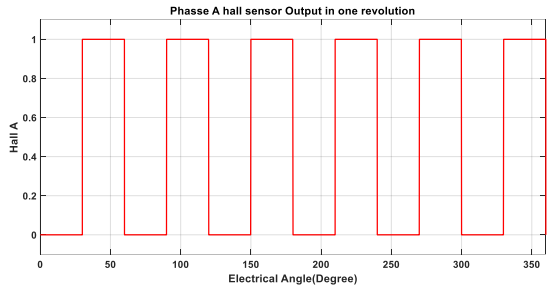


Fig. 7. hall sensor output graph at a full revolution of rotor in healthy condition

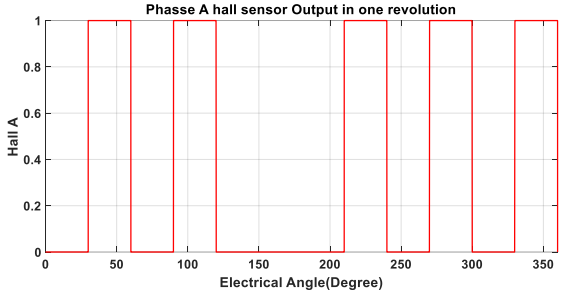


Fig. 8. hall sensor output graph at a full revolution of rotor in faulty condition

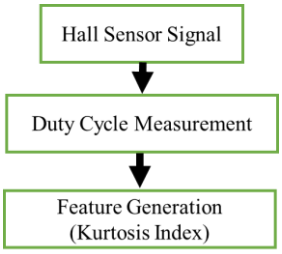


Fig. 9. BLDC motor demagnetization fault detection flowchart

For the classification of the hall sensor output in healthy and demagnetized conditions, the duty cycle feature can be used. Duty Cycle (DC) for a square-wave signal (like in Fig. 4) is defined as the ratio of the pulse width to the pulse period:

$$DC = \frac{Pulse\ Width}{Pulse\ Period} \tag{10}$$

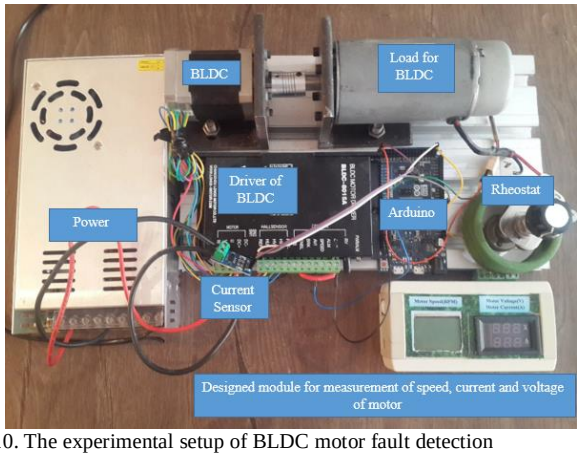


Fig. 10. The experimental setup of BLDC motor fault detection

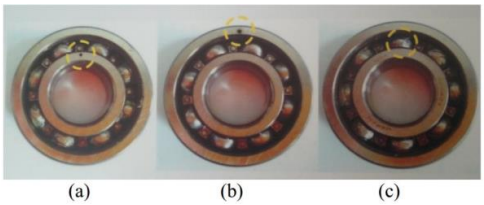


Fig. 11. hall sensor output graph at a full revolution of rotor in faulty condition

The pulse period can be measured by calculating the elapsed time between two successive falling edges, and pulse width can be measured by calculating the elapsed time between two successive rising edge and falling edge in Fig. 4. Therefore, in Fig. 7, it can be seen that the duty cycle is equal to 50 percent in all periods, while in Fig. 8 the duty cycles are not equal in all periods, and hence, in this condition some impulses will appear in the duty cycle measurement graph. Note that the duty cycle signal is measured online.

In order to compare the duty cycle signals in healthy or faulty conditions, kurtosis index can be used. Kurtosis is a mathematical representation that describes the impulsive features of time-domain signals. It is defined as the fourth standardized moment as:

$$Kurtosis = \frac{1}{N}\sum_{i=1}^{i=N}\left(\frac{x_i - \mu}{\sigma}\right)^4 \tag{11}$$

where x_i is the measured duty cycle signal in time domain, μ and σ are mean and variance of the duty cycle signal respectively, and N is the number of samples. Since the duty cycle signal in the faulty condition includes more impulsive features than the duty cycle in the healthy condition, kurtosis index can effectively classify the duty cycle signals. Finally, the flowchart of the proposed demagnetization fault detection is presented in Fig. 9.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Fig. 10 shows the experimental setup. In this setup, the BLDC motor speed is controlled in a dual loop speed-current closed-loop control using the PID controllers (as in Fig.1) and various signals such as current, voltage, vibration and speed can be measured. Using the ARDUINO hardware (model DUE) these signals are measured, and then transmitted to the

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MATLAB software (using serial port communication) where the proposed fault detection algorithms are carried.

B. Bearing Fault Detection

According to Fig. 11, various bearing defects (IRF, ORF, and BBF) are created. The BLDC motor speed, the decomposed speed and the CWT transform of the residual signal (Detailed signal) have been shown in Figs 12, 13, and 14. It is clear that in the healthy condition the amplitude of the residual signal is very low, and there is no significant symptom in the CWT transform of the residual signal. Therefore, it is concluded that the bearing is at a healthy condition.

In the case of IRF fault scenario, the motor speed, the decomposed speed and the CWT transform of the residual signal have been shown in Figs. 15, 16, and 17. It is clear that in this condition the amplitude of the residual signal has increased and some appreciable frequency peaks have appeared in the CWT transform of the residual signal. As shown in Fig. 17, if the reference command of the BLDC motor speed changes in every two seconds, then the residual frequency varies with time in every two seconds.

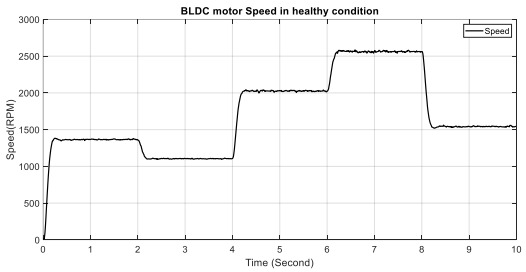


Fig. 12. BLDC motor speed signal in healthy condition

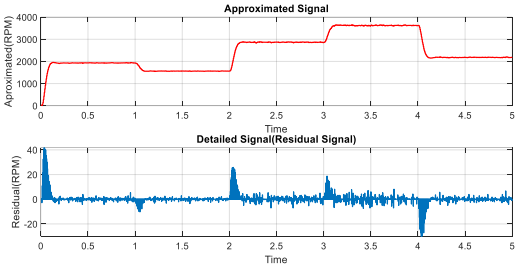


Fig. 13. Speed decomposition in healthy condition

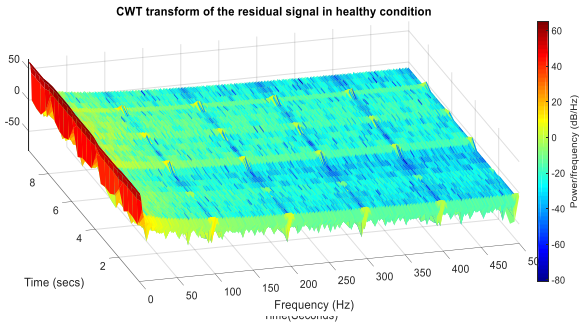


Fig. 14. Speed decomposition in IRF faulty condition

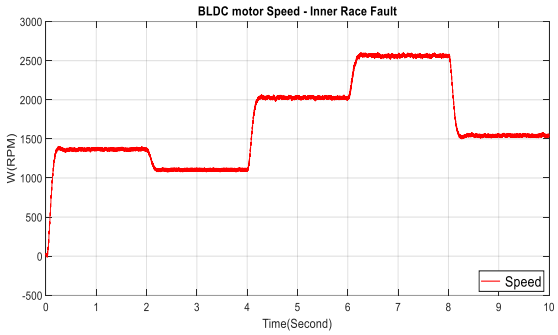


Fig. 15. BLDC motor speed signal in IRF faulty condition

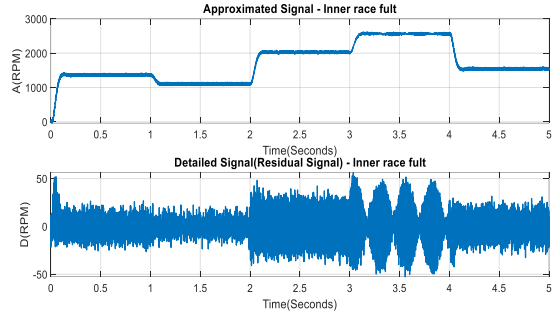


Fig. 16. Speed decomposition in IRF faulty condition

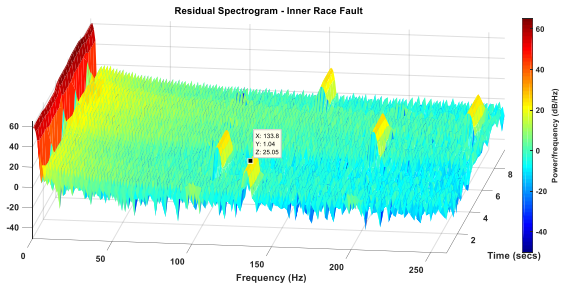


Fig. 17. CWT transform of the residual signal in IRF faulty condition

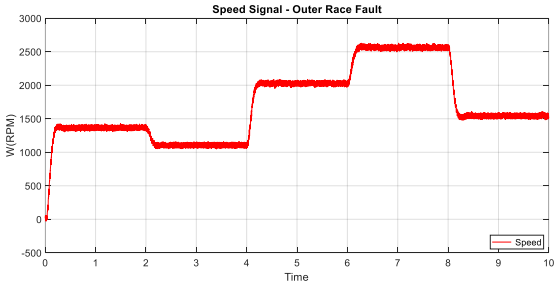


Fig. 18. BLDC motor speed signal in ORF faulty condition

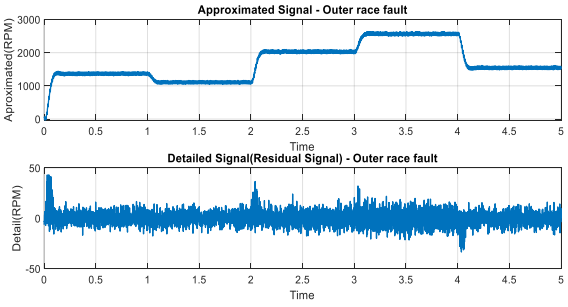


Fig. 19. Speed decomposition in ORF faulty condition

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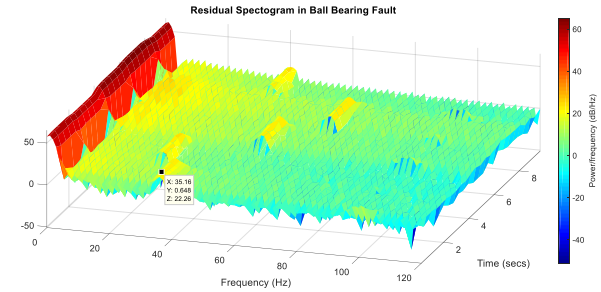


Fig. 20. CWT transform of the residual signal in ORF faulty condition

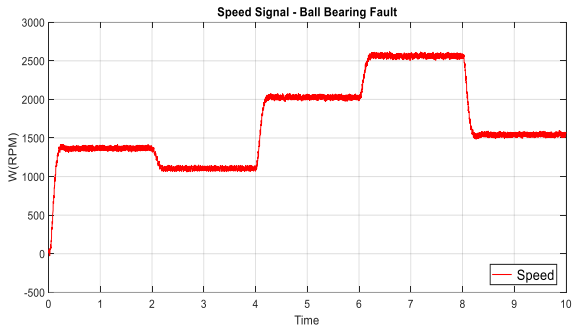


Fig. 21. BLDC motor speed signal in BBF faulty condition

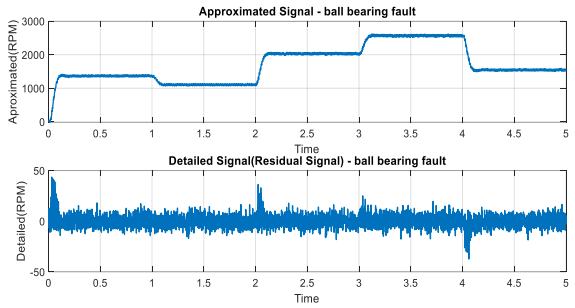


Fig. 22. Speed decomposition in BBF faulty condition

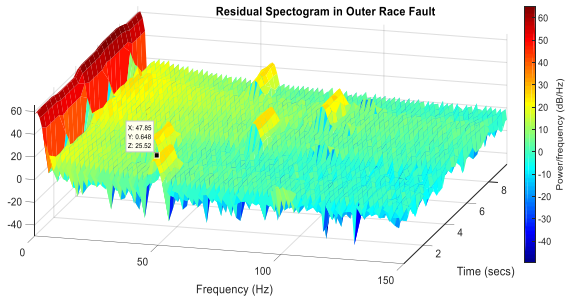


Fig. 23. CWT transform of the residual signal in ORF faulty condition



Fig. 24. Fully-demagnetized rotor

According to equation (1), and by substituting the bearing parameters, the relation between inner-race fault frequency and speed frequency is derived as $f_{in} = 5.7387 f_r$ which it is well consistent with the residual signal frequency peaks in Fig. 17. For example, in the first two seconds, the BLDC motor speed is equal to 1401 RPM which equivalent to a frequency of 23.35 Hz. By substituting this value in the above-mentioned equation, the inner race fault frequency is obtained equal to 134 Hz, on the other hand, according to Fig. 17, in the first two seconds the peak frequency is equal to 133.8 Hz. Therefore, the experimentally and theoretically computed frequencies are almost the same. Similar results are obtained for the outer-race and ball-bearing faults according to Figs. 18 to 23. To sum up, all the frequency peaks and their related motor average speed in Figs. 17, 20, and 23 have been summarized in Table I.

Also note that thanks to a large difference in the frequency contents of the detailed signal (arisen from bearing fault) and the approximated signal (arisen from reference variations in speed regulation of the BLDC motor), we can decompose these signals from speed signal only by using a single-level DWT decomposer. Also, the impulse response coefficients of g(n) and h(n) (in Fig. 3) were chosen based on the symmetric mother wavelet function (type 'sym1' in MATLAB).

TABLE I
FREQUENCY PEAKS AND THEIR RELATED MOTOR SPEED

Frequency Peak (Hz)	Figure Number	Motor Speed (RPM)
133.8 [0 2s]	15	1401
106.4 [2 4s]	15	1100
194.5 [4 6s]	15	2047
243.1 [6 8s]	15	2570
149.4 [8 10s]	15	1545
47.8 [0 2s]	18	1401
39.1 [2 4s]	18	1100
71.3 [4 6s]	18	2047
90.8 [6 8s]	18	2570
54.7 [8 10s]	18	1545
35.2 [0 2s]	21	1401
29.3 [2 4s]	21	1100
52.7 [4 6s]	21	2047
66.4 [6 8s]	21	2570
40.0 [8 10s]	21	1545

C. Demagnetization Fault Detection

In order to simulate the fully-demagnetized-rotor fault, one of the magnet bars shown in Fig. 24 was removed**.** Fig. 25 shows the BLDC motor speed and duty cycle signal in the healthy condition. It is clear that the duty cycle is almost equal to 50 percent in all the times. Therefore, it can be concluded that in the healthy condition the duty cycle is independent of speed. Now in Fig. 26, the BLDC motor speed and duty cycle signals in the fully demagnetized fault condition have been shown. It can be seen that in this condition, unlike the healthy condition, the duty cycle signal contains some periodic impulses because of the fully demagnetization. The kurtosis index of the duty cycle signal in the healthy and faulty condition is equal to 2.45 and 36.10, which shows that the kurtosis index in the faulty

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condition substantially differs from the kurtosis index in the healthy condition.

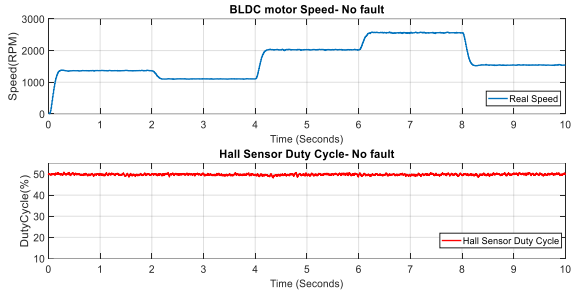


Fig. 25. Duty cycle signal in healthy condition

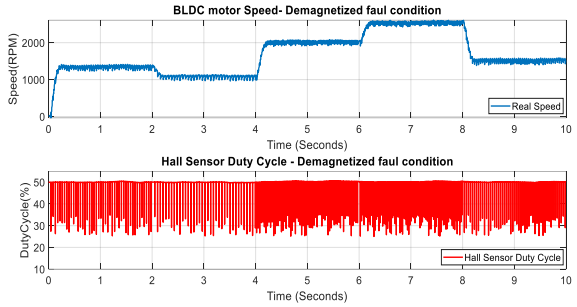


Fig. 26. Duty cycle signal in healthy condition

V. Conclusion

In this paper two novel algorithms for bearing and demagnetization fault detection in BLDC motors were presented. The proposed algorithms only use hall sensors which exist in the most BLDC motors. Therefore, since there is no need for redundant sensors such as accelerometers or current sensors, the proposed algorithms can be considered as a sensorless fault detection algorithm. At the first step, the effects of bearing and demagnetization faults on hall sensor outputs were studied. Then, using the cascade DWT-CWT transform of the speed signal, and the kurtosis index of the duty cycle signal of the hall sensor output, fault detection was performed and experimentally validated.

According to the experimentally results, bearing and demagnetization faults can be effectively detected and diagnosed using the proposed algorithms. Also some other advantages of the proposed algorithms include: firstly, due to the elimination of the redundant sensors, the cost of the diagnosis system is reduced significantly. Secondly, Because of the electrical isolation between the hall sensors and power circuits, the proposed algorithm has enough robustness against external noise and disturbances. Thirdly, the proposed algorithms work both in stationary and non-stationary operations of small and large power BLDC motors.

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